

AI vs Machine Learning vs Deep Learning

AI is the domain under which Machine Learning and Deep Learning are sub-topics.

Machine Learning is a subset of AI where systems learn patterns from data and make use of the learned patterns to determine the outputs, without being explicitly programmed to do so, which is what was done in traditional programming, where we made use of functions with a determined return type to give us the required output.

Data --> Learning Algorithm --> Model --> Prediction

Now let's say we need to predict an employee's hike, the data will consist of features like 'deliverables met', 'projects completed', 'hours spent', which the model will make use of to learn past patterns, and then give an output when someone asks it to predict the hike of an employee which the model previously had no idea about.

Deep Learning is a further subdivision of Machine Learning, that uses **neural networks** to learn complex patterns from data, which simple ML models fail to do.

It works on the same principle of Machine Learning but is suited for complex, often unstructured data such as images, videos, audio files, and other forms of multimodal data.

We also have additional terminologies which you guys might be familiar with,

Generative AI which is the most commonly used type of AI, which learns patterns from existing data, and then proceeds to generate new content. Text generation, Image generation, Code generation are few of the use cases of generative AI.

Reinforcement Learning is a relatively obscure terminology, which makes use of a reward/penalty system to train models. The model interacts with an environment, and depending on its decisions its either rewarded, or punished.

Suppose we are training a model to play a game. If it takes a particular move which causes it to lose, we give it -5 as a penalty. This motivates the model to **not** take that particular move, and instead look for a move, which hopefully gives it a reward. This type of learning is most common in robotic systems.